STUDY NOTE · MATHEMATICS OF ML

The Kalman Filter from First Principles

1 · The model

Linear Gaussian state-space: $x_t = A x_{t-1} + w_t$, $z_t = H x_t + v_t$,
with $w \sim N(0, Q)$, $v \sim N(0, R)$. The filter maintains $p(x_t \mid z_{1:t}) = N(\hat{x}_t, P_t)$.

2 · Prediction step

$\hat{x}_{t|t-1} = A \hat{x}_{t-1}$, $P_{t|t-1} = A P_{t-1} A^T + Q$

3 · Update (correction) step

Innovation: $y_t = z_t - H \hat{x}_{t|t-1}$
Kalman gain: $K_t = P_{t|t-1} H^T (H P_{t|t-1} H^T + R)^{-1}$
Posterior: $\hat{x}_t = \hat{x}_{t|t-1} + K_t y_t$
$P_t = (I - K_t H) P_{t|t-1}$

4 · Why it works

The Kalman gain is the MAP/MMSE estimator: it weighs model vs. measurement by their covariance - a theme shared with Approximate Message Passing, where beliefs play the role of ($\hat{x}$, P) pairs. Hover over the interactive derivations on the web version for step-by-step algebra.

Full interactive version with LaTeX, diagrams and a runnable notebook:
yousof.example.com/notes/kalman-filter-notes